

The experimental objective in June:

Module 1: Verification of BW-pYB1a-*vdh-nox*, HPLC of BW-pYB1a-*vdh-nox* and validation of single-gene expression by protein analysis

Module 2: To construct the ODM random-mutation library by error-prone PCR, and characterize the mCherry-based 4-HB reporter together with the chloramphenicol detection system.

Module 3: To construct recombinant plasmids of the vanillic acid degradation pathway, verify target protein expression by SDS-PAGE, and validate the full pathway function using a pyruvate sensor strain.

Week 3 (5.28-6.4)

Module 1:

1. Inoculation

Original strains: BW-pYB1a-*vdh*、BW-pYB1a-*progo*、BW-pYB1a-*nov1f*

Cultivation in a shaker at 37°C for 12 hours

2. Induction

Induction of BW-pYB1a-*vdh*、BW-pYB1a-*progo*、BW-pYB1a-*nov1f*

Induction System Component	Volume
ZY	4.8 mL
50x 5052 supplement	100 μ L
50x M supplement	100 μ L
MgSO ₄	10 μ L
1000x trace element	10 μ L
Bacterial suspension	50 μ L
arabinose	50 μ L
Antibiotic	5 μ L

Cultivation in a shaker at 30°C for 16 hours.

3. Protein expression

1. Sample Preparation

The pYB1a-*Vdh*、pYB1a-*Progo*、pYB1a-*Nov1f* strains were cultured under induction conditions at 30°C for 16 hours. The cultures were collected, and the bacterial suspension with a OD₆₀₀=6 unit was measured to obtain the final sample. After centrifugation at 10,000 g for 10 minutes, the supernatant was discarded, and the precipitate was resuspended in 600 μ L sterile water. The cells were lysed using an ultrasonic cell disruptor. The lysate was then centrifuged at 10,000 g for 10 minutes to separate the supernatant and sediment. Finally, the precipitate was resuspended in 600

μL sterile water. This procedure completed the sample preparation.

2.point sample

Prepare the protein buffer solution according to the table below. Place the gel in the electrophoresis instrument. Transfer the sample into the PCR instrument and heat at 100°C for 10 minutes to denature proteins. After checking for leaks with the buffer solution, mix $32\ \mu\text{L}$ of prepared sample with $8\ \mu\text{L}$ loading buffer. After that, load $12\ \mu\text{L}$ of the mixture into the gel well and add $10\ \mu\text{L}$ of protein marker. The sample loading is now complete.

Concentrating Gel:

12% Lower Gel	2 Gel Units	3 Gel Units	4 Gel Units	5 Gel Units	6 Gel Units	8 Gel Units	10 Gel Units
Solution Components (mL)	10	15	20	25	30	40	50
Water	3.3	4.9	6.6	8.2	9.9	13.2	16.5
30% Acrylamide Solution	4	6	8	10	12	16	20
Tris (pH 8.8)	2.5	3.8	5	6.3	7.5	10	12.5
10% SDS	0.1	0.15	0.2	0.25	0.3	0.4	0.5
10% Ammonium Persulfate	0.1	0.15	0.2	0.25	0.3	0.4	0.5
TEMED	0.004	0.006	0.008	0.01	0.012	0.016	0.02

separation gel:

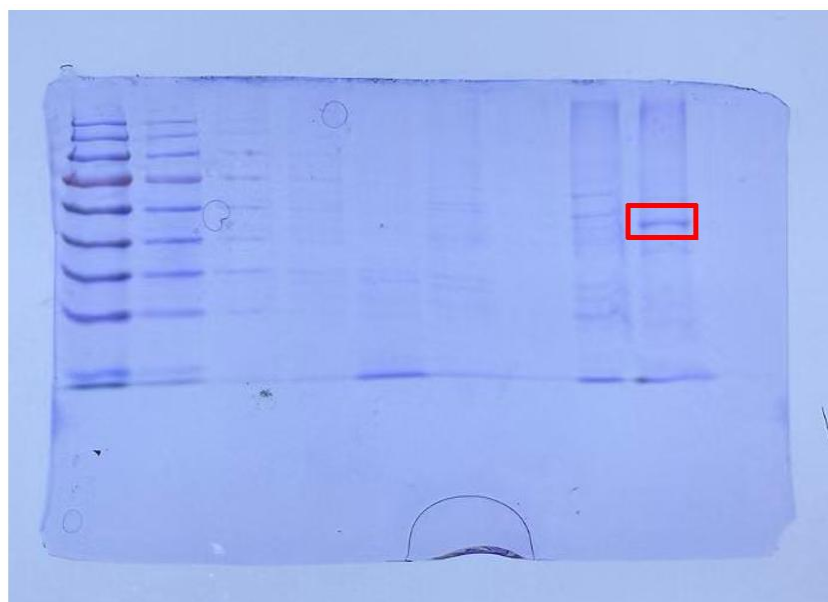
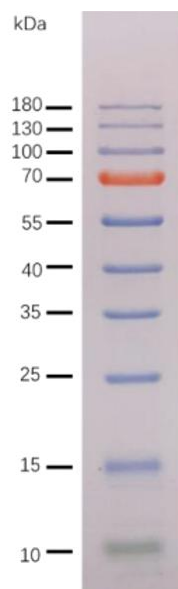
5% Stacking Gel	2 Gel Units	3 Gel Units	4 Gel Units	5 Gel Units	6 Gel Units	8 Gel Units	10 Gel Units	12 Gel Units
Solution Components (mL)	2	3	4	5	6	8	10	12
ddH ₂ O	1.4	2.1	2.7	3.4	4.1	5.5	6.8	8.1
30% Acrylamide Solution	0.33	0.5	0.67	0.83	1	1.3	1.7	2.01
Tris (pH 8.8)	0.25	0.38	0.5	0.63	0.75	1	1.25	1.5
10% SDS	0.02	0.03	0.04	0.05	0.06	0.08	0.1	0.12
10% Ammonium Persulfate (APS)	0.02	0.03	0.04	0.05	0.06	0.08	0.1	0.12
TEMED	0.002	0.003	0.004	0.005	0.006	0.008	0.01	0.012

3.electrophoresis

Connect the power supply and run the gel at 80 V for 30 minutes, then run it at 120 V for about 1 hour

4.Dyeing and decoloring

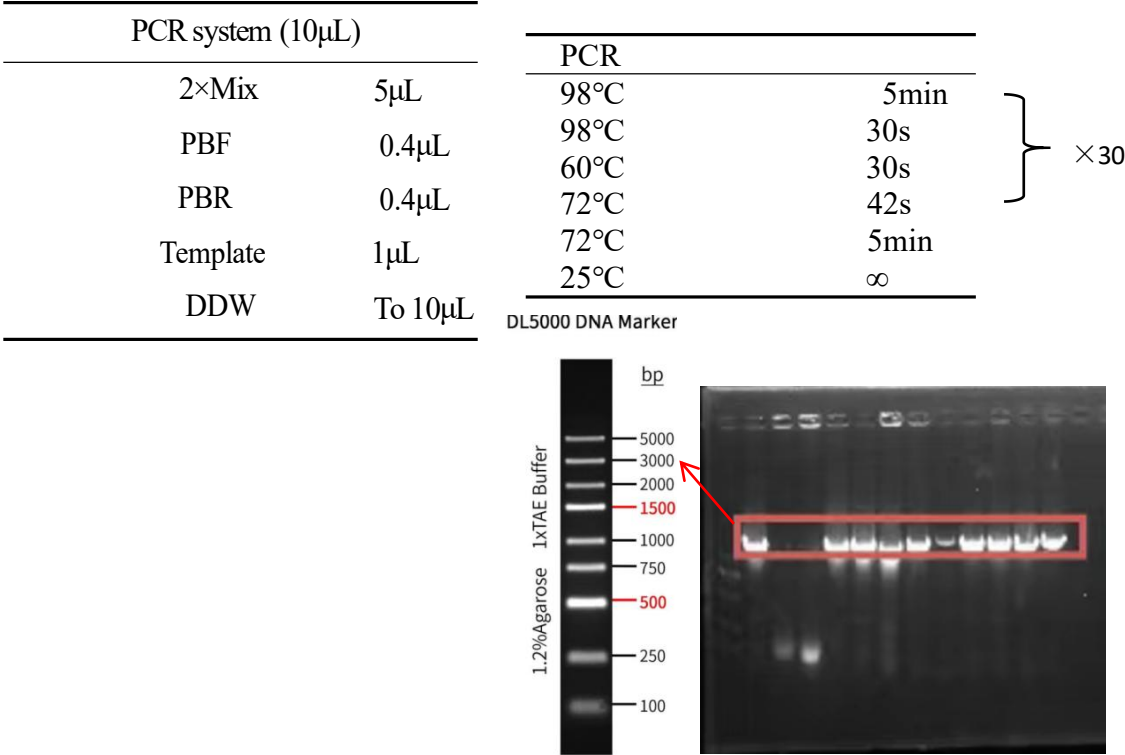
Gently remove the gel from the water, then place it in the staining solution. Boil the solution and incubate the gel with heated staining liquid on a small shaker at room temperature for approximately 45 minutes. Rinse the gel thoroughly with distilled water to remove residual staining. Add protein decolorizing solution, heat it to boiling, and decolorize the heated solution on a small shaker at room temperature for 30 minutes. The results are shown in the figures:



The results showed that the protein was well soluble and could be expressed normally in cells after 30°C induction for 20h.

4. Colony PCR of BW-pYB1a-*vdh-nox*

PBF/PBR was chosen as the primer with a theoretical length of 3216 bp



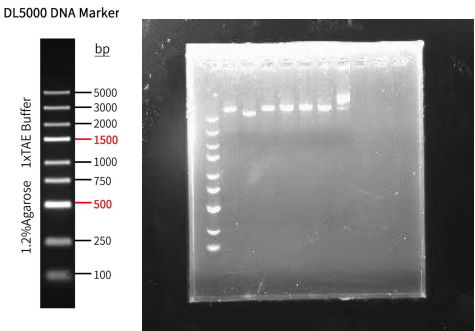
The colony PCR is successful.

5. Enzyme digestion

Selected 10 colonies for using SpeI PstI to verify the BW-pYB1a-*vdh-nox*. Enzyme digestion system:

Enzyme digestion (10μL)	
SpeI	0.2μL
PstI	0.2μL
rCutsmart	1.0μL
pYB1a- <i>vdh-nox</i>	200ng
DDW	to 10μL

The products were analyzed by agarose gel electrophoresis. The results showed that the enzyme digestion was failure.



Module 2:

(A)Fifth- round error- prone PCR

1. Error-prone PCR

Error-prone PCR was performed with two different Mg^{2+}/Mn^{2+} ratios to introduce random mutations into ODM coding sequence.

$Mg^{2+} : Mn^{2+} = 10:1$

PCR system (50 μ L)	
template	1ng
F	2ul
R	2ul
10 mM Mg^{2+} solution	1.5 μ L
1 mM Mn^{2+} solution	1.5 μ L
Green Mix	25 μ L
DDW	17 μ L

$Mg^{2+} : Mn^{2+} = 5:1$

PCR system (50 μ L)	
template	1ng
F	2ul
R	2ul
10 mM Mg^{2+} solution	1.5 μ L
1 mM Mn^{2+} solution	3 μ L
Green Mix	25 μ L
DDW	15.5 μ L

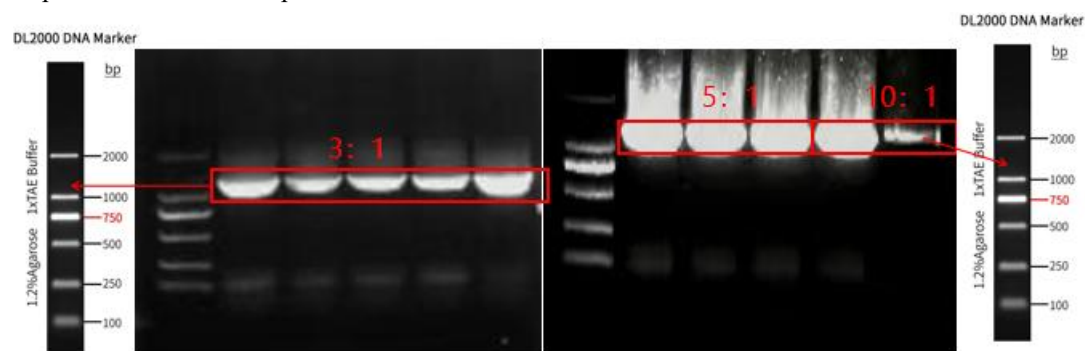
$Mg^{2+} : Mn^{2+} = 3:1$

PCR system (50 μ L)	
template	1ng
F	2ul
R	2ul
10 mM Mg^{2+} solution	1.94 μ L
1 mM Mn^{2+} solution	6.45 μ L
Green Mix	25 μ L
DDW	11.61 μ L

PCR		
98°C	5min	} $\times 35$
98°C	30s	
56°C	30s	
72°C	45s	
72°C	5min	
25°C	∞	

2. Conducted gel extraction for the fragments.

Expected band size: 1133 bp



Partial smearing was observed in the 5:1 and 10:1 groups, presumably due to excessive sample loading into the wells. Solution: Reduce the sample loading volume for each well.

The gel-extracted fragment exhibited the correct target band.

10: 1体系	5: 1体系	3: 1体系
151ng/ μ L	212.5ng/ μ L	129.4ng/ μ L

2. Connection

Connecting the two fragments using the Golden-Gate method

Goden-gate system (10μL) 5:1	10: 1	
template	200ng	100ng
10×T4 buffer	1 μL	1 μL
BsaI	0.5 μL	0.5 μL
T4 DNA ligase	0.3 μL	0.3 μL
rCutSmart (10×)	1μL	1μL
vector	1 μL	1 μL
DDW	10 μL	10 μL

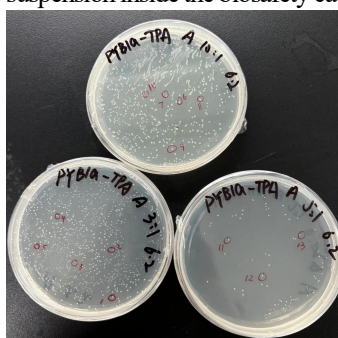
Connection		
temperature	time	cycles
37°C	30min	
37°C	3min	25 X
16°C	3 min	25 X
16°C	20min	
37°C	20min	
55°C	15min	
80°C	15min	
25°C	∞	

3.Transformation

The ligated system was transferred into DH5α receptor cells, coated with plates, and cultured inverted at 37°C for 12h;

4.Plating results

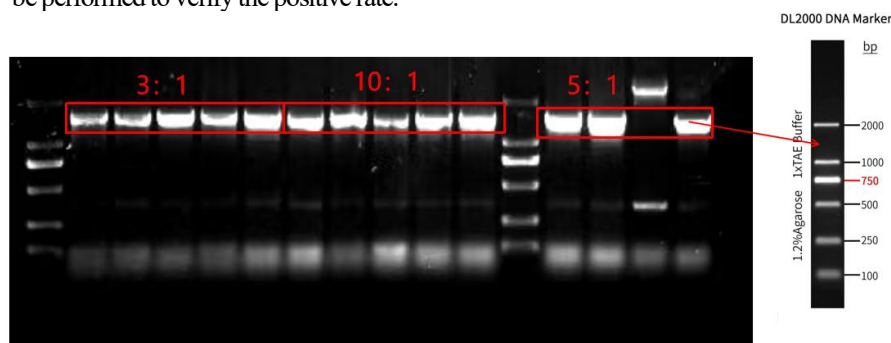
The 5:1 group showed low library capacity, which may be attributed to incorrect ligation. Since competent cells used for the three plates were prepared at different time points, low competent cell efficiency is another suspected cause. No smeared growth was observed this time. The improved protocol: aspirate 100 μL of the bacterial suspension inside the biosafety cabinet to reduce cell concentration, followed by centrifugation and plating.



5.Colony PCR

PBF/PBR was chosen as the primer with a theoretical length of 1508bp.

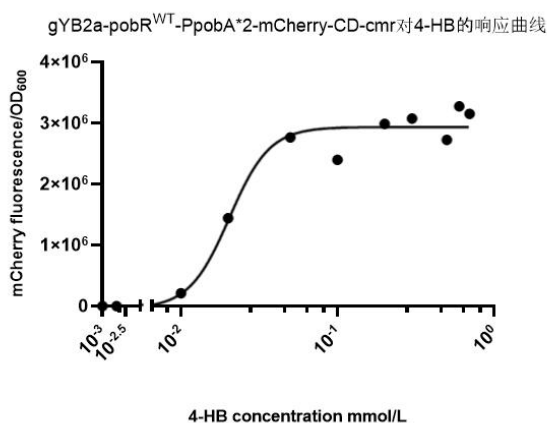
Colony-PCR of the 3:1 and 10:1 groups yielded correct target bands, indicating successful ligation. One band from the 5:1 group was incorrect, so the positive rate was not 100%. Further colony-PCR of more colonies will be performed to verify the positive rate.



(B) Curve of mCherry

1.Final curve of mCherry

The 4-HB concentration gradient was set as 0, 0.001, 0.002, 0.008, 0.01, 0.02, 0.05, 0.1, 0.3, 0.4, 0.5, 0.7, 0.8 mmol/L.



(C) Chloramphenicol detection curve

1.system

Basal system (no 4-HB): 1 μ L bacterial culture; 0.2 μ L Amp stock. Total volume of 200 μ L per well containing 4-HB, LB medium and chloramphenicol at varying concentrations. A control without chloramphenicol was additionally set up. The optimal stable concentration of 4-HB was fixed at 0.6 mmol/L. The 4-HB stock solution was diluted from 100 mmol/L to 10 mmol/L. Chloramphenicol stock solution was prepared at 0.3 mg/mL. The chloramphenicol gradient was designed as 0, 3, 7.5, 15, 22.5, 30, 37.5, 45, 52.5 mg/L. Incubate at 37 °C for 10 h, with measurements taken every 2 h.

96- well plate setup:

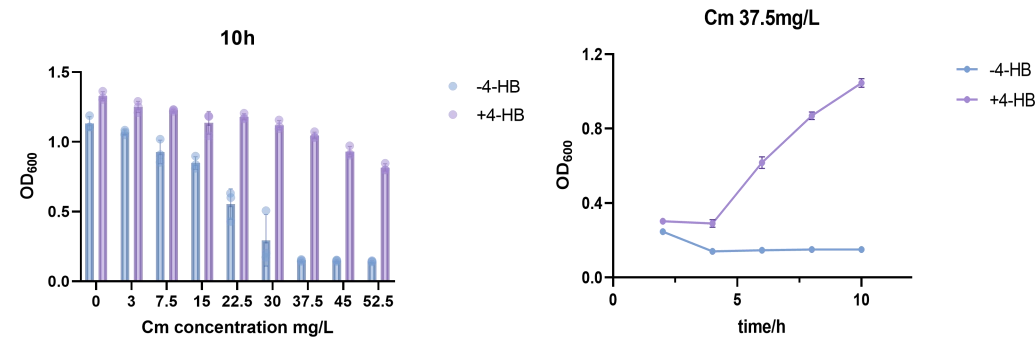
No.	Cm concentration gradient mg/L	4-HB(0.6mmol/L)			non-4-HB	
		Cm(μ l)	4-HB(μ l)	LB(μ l)	Cm(μ l)	LB(μ l)
1	0	0	12	188	0	200
2	3	2	12	186	2	198
3	7.5	5	12	183	5	195
4	15	10	12	178	10	190
5	22.5	15	12	173	15	185
6	30	20	12	168	20	180
7	37.5	25	12	163	25	175
8	45	30	12	158	30	170
9	52.5	35	12	153	35	165

OD values were measured at 2, 4, 6, 8 and 10 hours of incubation.

2.OD measurement

The difference reached its maximum at the 10th hour, indicating that the optimal detection window was missed due to excessive incubation time in the previous experiment. Growth curves of bacteria were measured under chloramphenicol concentrations of 22.5, 30 and 37.5 mg/L. The maximum difference in bacterial growth was observed at 37.5 mg/L. However, the groups supplemented with 4-HB did not reach the plateau phase. Further incubation for 12 h will be carried out to explore the plateau phase.

OD values were measured at 2, 4, 6, 8 and 10 hours of incubation.



Module 3:

(A) Construct BW-pLB1s-

(B) *AB-aroY*, BW-pYB1a-*dmpB-dmpD-mhpD-mhpE*

1.Colony PCR verification

Single colonies were picked as templates for colony PCR. Colony PCR system (10μL):
Colony PCR system (10μL):

Green Mix	5μL
Primer-F	0.4μL
Primer-R	0.4μL
DDW	4.2μL
Colony	—

PCR program:

98°C	5min	} X30
98°C	30s	
57°C	30s	
72°C	2min	
72°C	5min	
25°C	∞	

The PCR products were verified by agarose gel electrophoresis.

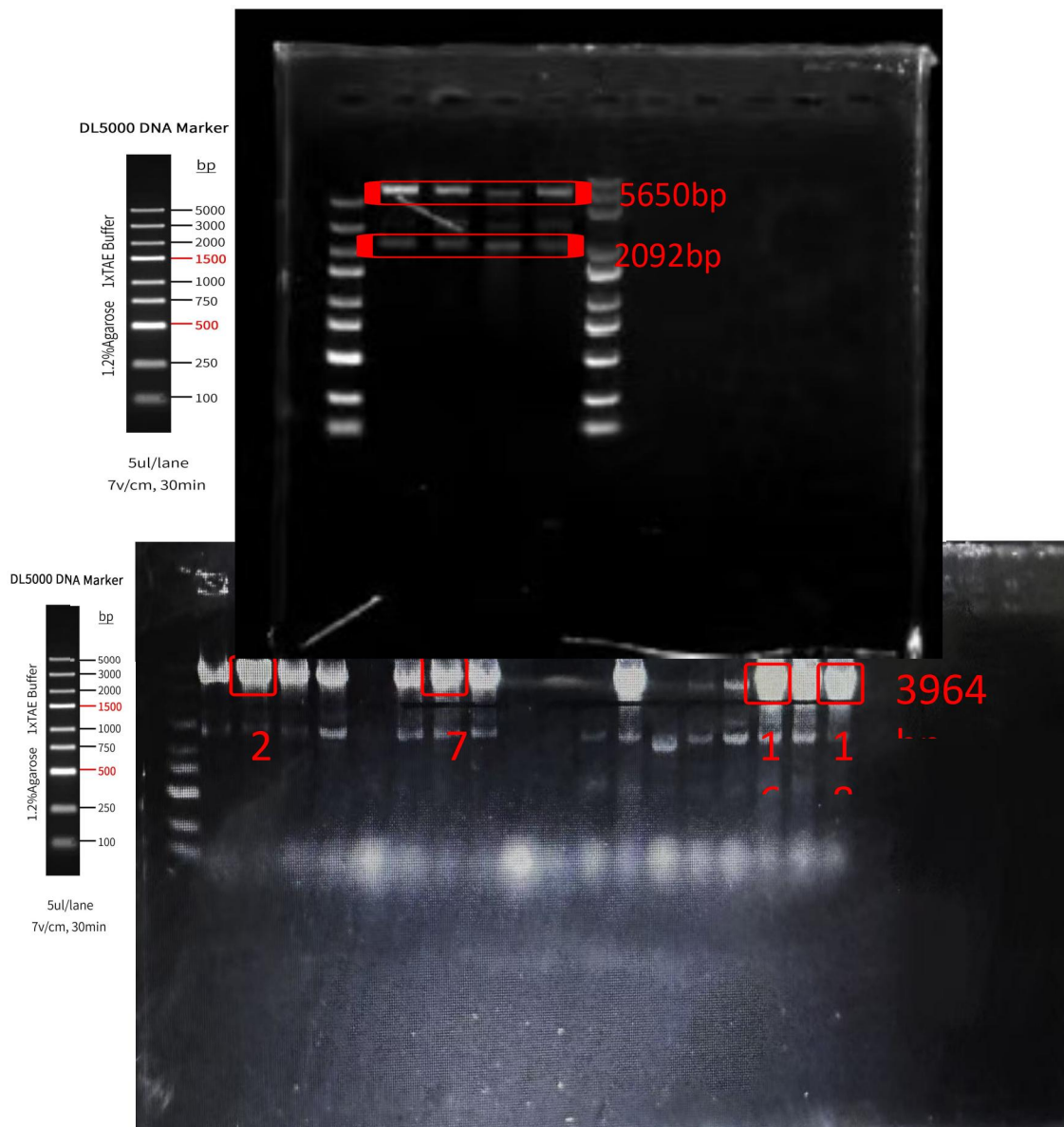
2.Ligation, transformation and plating

Since the sequencing result at the end of last month showed that the gene was not successfully inserted, the ligation, transformation and plating were continued this week. The recovered fragments were ligated by Gibson assembly:

Gibson system (10 μ L):

c117	5 μ L
Fragment	100ng
Vector	100ng
DDW	to 10 μ L

Gibson: 50°C 15min; 4°C ∞ . The ligation product was transformed into DH5 α competent cells, plated, and cultivated at 37°C for 12 hours.



3.Shake culture

50 μ L of the bacterial preservation solution was transferred to 5mL of liquid LB medium and cultivated in a shaking incubator at 37°C for 12 hours.

4.Plasmid extraction

After cultivation, plasmids were extracted and the concentrations were measured.

5.Enzyme digestion

The extracted plasmids were digested for verification.

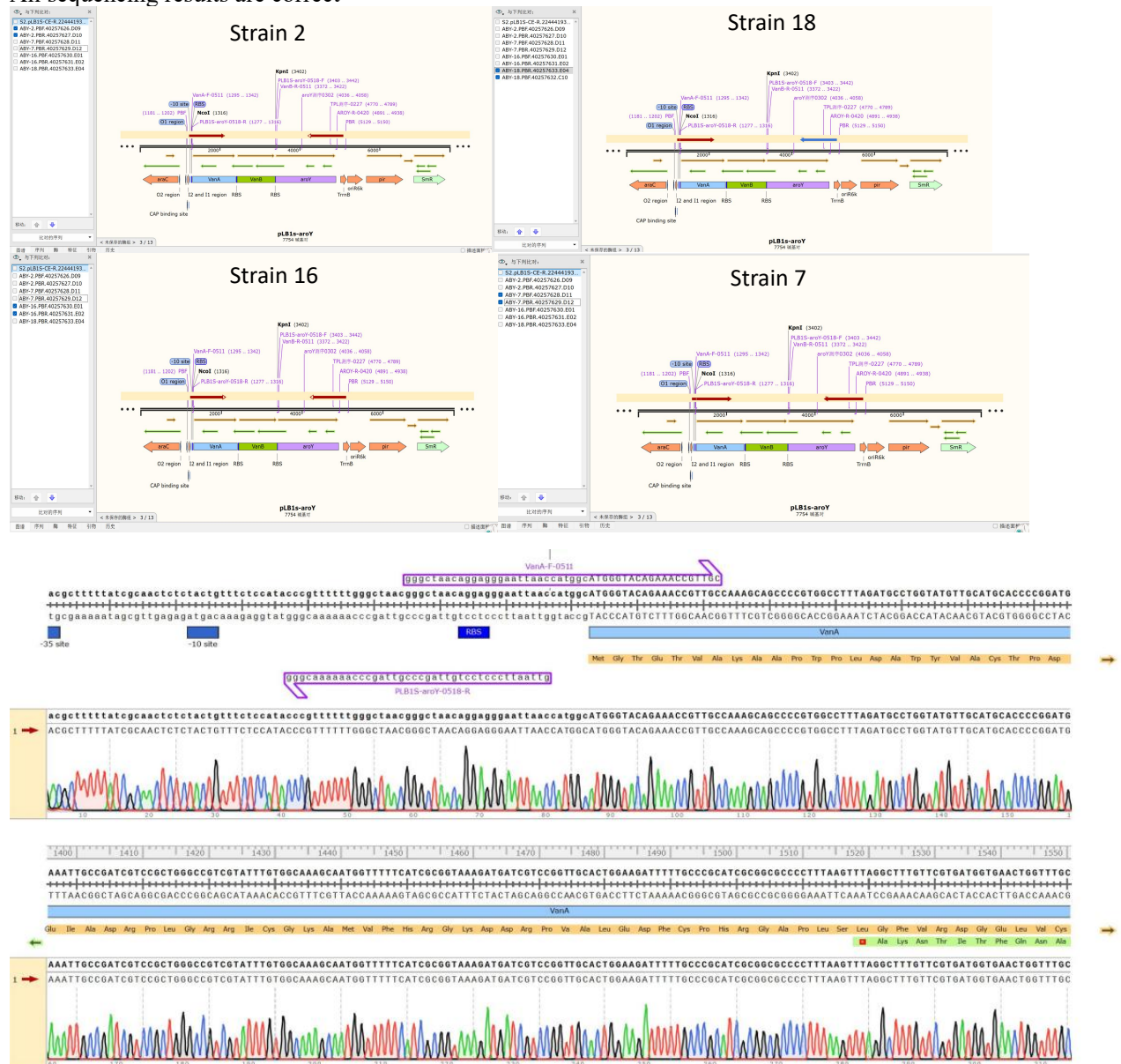
6. Transformation (*VanAB-aroY* and *BDDE*)

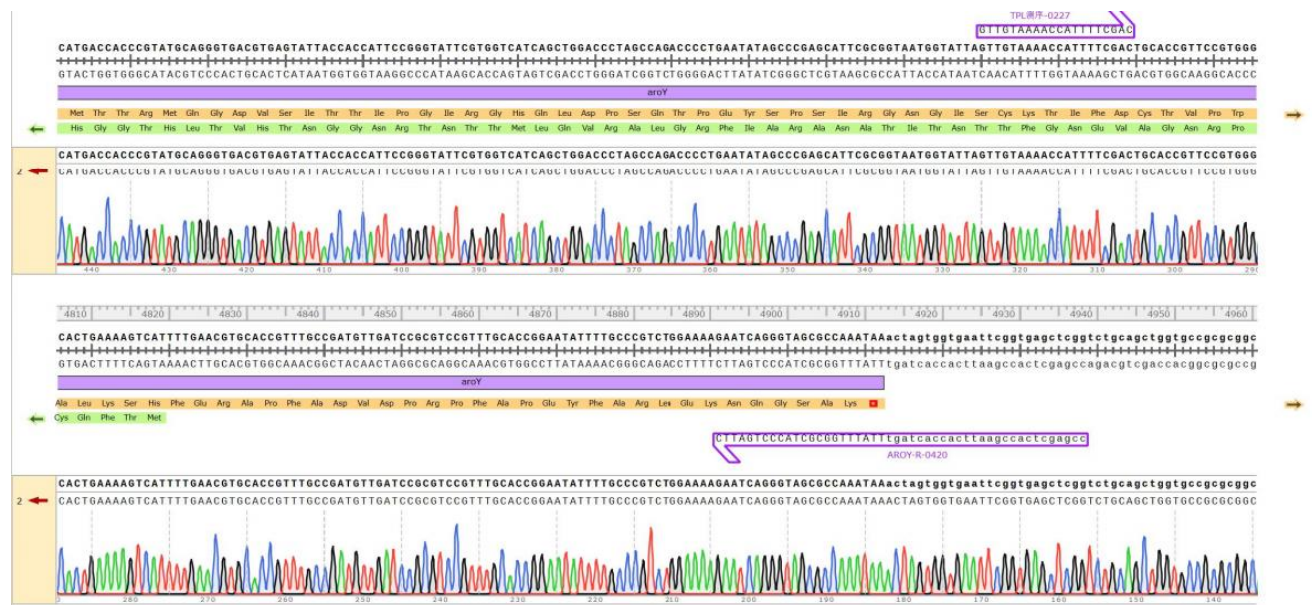
The *VanAB-aroY* (pLB1s) and *BDDE* (pYB1a-*dmpB-dmpD-mhpD-mhpE*) constructs were transformed into competent cells, plated, and cultivated at 37°C for 12 hours.

7. Send plasmids for sequencing

The constructed or verified plasmids were submitted to the sequencing company for verification.

All sequencing results are correct





(B) Validation of single-gene expression by protein analysis

1. Shake culture in LB medium.

50μL of the bacterial preservation solution was transferred to 5mL of liquid LB medium and

2. Induction.

50μL of bacterial fluid was taken and transferred to the ZY induction system, and cultivated in a shaking incubator at 30°C for 16 hours. The induction system is as follows:

ZY induction system:

Component	Volume
ZY	4.8mL
50×5052 supplement	100μL
50×M supplement	100μL
MgSO ₄	10μL
1000× trace element	10μL
Bacterial suspension	50μL
arabinose	50μL
Antibiotic	5μL

2. Sample Preparation

The cultures were collected, and the bacterial suspension with a OD₆₀₀=6 unit was measured to obtain the final sample. After centrifugation at 10,000 g for 10 minutes, the supernatant was discarded, and the precipitate was resuspended in 600 μL sterile water. The cells were lysed using an ultrasonic cell disruptor. The lysate was then centrifuged at 10,000 g for 10 minutes to separate the supernatant and sediment. Finally, the precipitate was resuspended in 600 μL sterile water.

This procedure completed the sample preparation.

3. point sample

Prepare the protein buffer solution according to the table below. Place the gel in the electrophoresis instrument. After checking for leaks with the buffer solution, mix 16 μL of prepared sample with 4 μL loading buffer. Transfer the mixed sample into the PCR instrument and heat at 100°C for 5 minutes to denature proteins. After cooling, load 16 μL of the mixture into the gel well and add 4 μL of protein marker. The sample loading is now complete.

Concentrating Gel:

	2 Gel Units	3 Gel Units	4 Gel Units	5 Gel Units	6 Gel Units	8 Gel Units	10 Gel Units	12 Gel Units
5% Stacking Gel								
Solution Components (mL)	2	3	4	5	6	8	10	12
ddH ₂ O	1.4	2.1	2.7	3.4	4.1	5.5	6.8	8.1
30% Acrylamide Solution	0.33	0.5	0.67	0.83	1	1.3	1.7	2.01
Tris (pH 8.8)	0.25	0.38	0.5	0.63	0.75	1	1.25	1.5
10% SDS	0.02	0.03	0.04	0.05	0.06	0.08	0.1	0.12
10% Ammonium Persulfate (APS)	0.02	0.03	0.04	0.05	0.06	0.08	0.1	0.12
TEMED	0.002	0.003	0.004	0.005	0.006	0.008	0.01	0.012

separation gel:

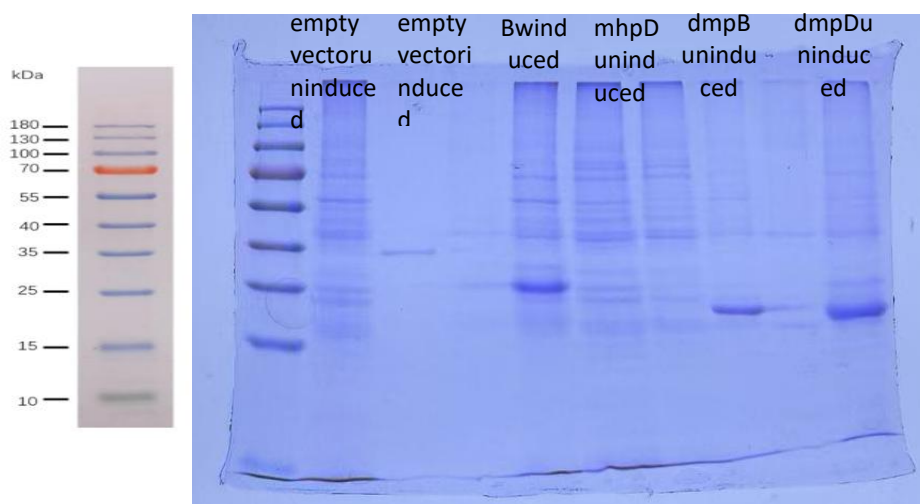
12% Lower Gel	2 Gel Units	3 Gel Units	4 Gel Units	5 Gel Units	6 Gel Units	8 Gel Units	10 Gel Units
Solution Components (mL)	10	15	20	25	30	40	50
Water	3.3	4.9	6.6	8.2	9.9	13.2	16.5
30% Acrylamide Solution	4	6	8	10	12	16	20
Tris (pH 8.8)	2.5	3.8	5	6.3	7.5	10	12.5
10% SDS	0.1	0.15	0.2	0.25	0.3	0.4	0.5
10% Ammonium Persulfate	0.1	0.15	0.2	0.25	0.3	0.4	0.5
TEMED	0.004	0.006	0.008	0.01	0.012	0.016	0.02

4.electrophoresis

Connect the power supply and run the gel at 80 V for 30 minutes, then run it at 120 V for about 1 hour

5. Dyeing and decoloring

Gently remove the gel from the water, then place it in the staining solution. Boil the solution and incubate the gel with heated staining liquid on a small shaker at room temperature for approximately 45 minutes. Rinse the gel thoroughly with distilled water to remove residual staining. Add protein decolorizing solution, heat it to boiling, and decolorize the heated solution on a small shaker at room temperature for 30 minutes. The results are shown in the figures:



Week 4 (6.5-6.11)

Module 1:

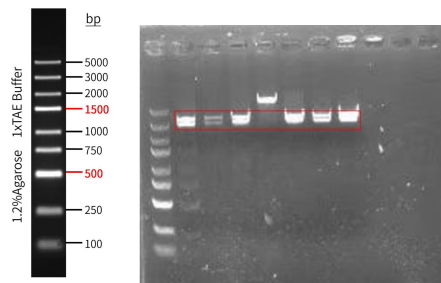
(A)Verification of BW-pYB1a-*vdh-nox*

1.Enzyme digestion

Selected 10 colonies for using *NocI* *PstI* to verify the BW-pYB1a-*vdh-nox*. Enzyme digestion system:

Enzyme digestion (10μL)	
<i>NocI</i>	0.2μL
<i>PstI</i>	0.2μL
rCutsmart	1.0μL
pYB1a- <i>vdh-nox</i>	200ng
DDW	to 10μL

The products were analyzed by agarose gel electrophoresis. The results showed that the enzyme digestion was failure.



(B) Validation of single-gene expression by protein analysis

1. Inoculation

Original strains: BW-pYB1a-*vdh*, BW-pYB1a-*progo*, BW-pYB1a-*nox*, BW-pYB1a-*novlf*, BW25113, BW-pYB1a-eGFP

Cultivation in a shaker at 37°C for 12 hours

2. Induction

Induction of BW-pYB1a-*vdh*, BW-pYB1a-*progo*, BW-pYB1a-*nox*, BW-pYB1a-*novlf*, BW25113, BW-pYB1a-eGFP

Induction System Component	Volume
ZY	4.8 mL
50x 5052 supplement	100 μ L
50x M supplement	100 μ L
MgSO ₄	10 μ L
1000x trace element	10 μ L
Bacterial suspension	50 μ L
arabinose	50 μ L
Antibiotic	5 μ L

Cultivation in a shaker at 30°C for 16 hours.

3. protein detection

1. Sample Preparation

The pYB1a-Vdh-Nox, pYB1a-Vdh Protein strains were cultured under induction conditions at 30 °C for 16 hours. The cultures were collected, and the bacterial suspension with a OD₆₀₀=6 unit was measured to obtain the final sample. After centrifugation at 10,000 g for 10 minutes, the supernatant was discarded, and the precipitate was resuspended in 600 μ L sterile water. The cells were lysed using an ultrasonic cell disruptor. The lysate was then centrifuged at 10,000 g for 10 minutes to separate the supernatant and sediment. Finally, the precipitate was resuspended in 600 μ L sterile water. This procedure completed the sample preparation.

2. point sample

Prepare the protein buffer solution according to the table below. Place the gel in the electrophoresis instrument. Transfer the sample into the PCR instrument and heat at 100 °C for 10 minutes to denature proteins. After checking for leaks with the buffer solution, mix 32 μ L of prepared sample with 8 μ L loading buffer. After that, load 12 μ L of the mixture into the gel well and add 10 μ L of protein marker. The sample loading is now complete.

Concentrating Gel:

12% Lower Gel	2 Gel Units	3 Gel Units	4 Gel Units	5 Gel Units	6 Gel Units	8 Gel Units	10 Gel Units
Solution Components (mL)	10	15	20	25	30	40	50
Water	3.3	4.9	6.6	8.2	9.9	13.2	16.5
30% Acrylamide Solution	4	6	8	10	12	16	20
Tris (pH 8.8)	2.5	3.8	5	6.3	7.5	10	12.5
10% SDS	0.1	0.15	0.2	0.25	0.3	0.4	0.5
10% Ammonium Persulfate	0.1	0.15	0.2	0.25	0.3	0.4	0.5
TEMED	0.004	0.006	0.008	0.01	0.012	0.016	0.02

separation gel:

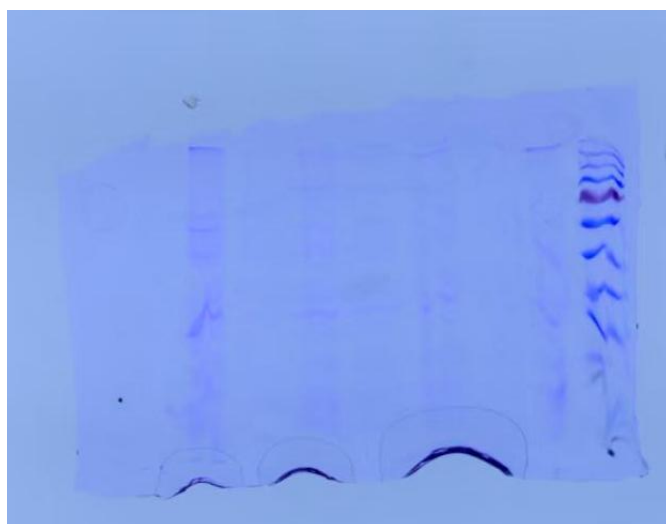
5% Stacking Gel	2 Gel Units	3 Gel Units	4 Gel Units	5 Gel Units	6 Gel Units	8 Gel Units	10 Gel Units	12 Gel Units
Solution Components (mL)	2	3	4	5	6	8	10	12
ddH ₂ O	1.4	2.1	2.7	3.4	4.1	5.5	6.8	8.1
30% Acrylamide Solution	0.33	0.5	0.67	0.83	1	1.3	1.7	2.01
Tris (pH 8.8)	0.25	0.38	0.5	0.63	0.75	1	1.25	1.5
10% SDS	0.02	0.03	0.04	0.05	0.06	0.08	0.1	0.12
10% Ammonium Persulfate (APS)	0.02	0.03	0.04	0.05	0.06	0.08	0.1	0.12
TEMED	0.002	0.003	0.004	0.005	0.006	0.008	0.01	0.012

3. electrophoresis

Connect the power supply and run the gel at 80 V for 30 minutes, then run it at 120 V for about 1 hour

4. Dyeing and decoloring

Gently remove the gel from the water, then place it in the staining solution. Boil the solution and incubate the gel with heated staining liquid on a small shaker at room temperature for approximately 45 minutes. Rinse the gel thoroughly with distilled water to remove residual staining. Add protein decolorizing solution, heat it to boiling, and decolorize the heated solution on a small shaker at room temperature for 30 minutes. The results are shown in the figures:



Poor gel preparation and insufficient protein loading lead to invisible bands in SDS-PAGE. Buffer leakage during electrophoresis causes distorted bands.

Module 2:

(A) Sixth- round error- prone PCR amplification of ODM gene

Error-prone PCR was performed with two different Mg²⁺/Mn²⁺ ratios to introduce random mutations into ODM coding sequence.



PCR system (50μL)



PCR system (50μL)

template	1ng
F	2ul
R	2ul
10 mM Mg ²⁺ solution	1.5μL
1 mM Mn ²⁺ solution	1.5μL
Green Mix	25μL
DDW	17μL

Mg²⁺ : Mn²⁺ = 3:1

PCR system (50μL)	
template	1ng
F	2ul
R	2ul
10 mM Mg ²⁺ solution	1.94μL
1 mM Mn ²⁺ solution	6.45μL
Green Mix	25μL
DDW	11.61μL

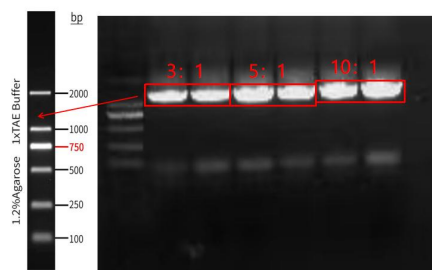
PCR	
98°C	5min
98°C	30s
66°C	30s
72°C	45s
72°C	5min
25°C	∞

} × 35

2. Conducted gel extraction for the fragments.

First gel extraction

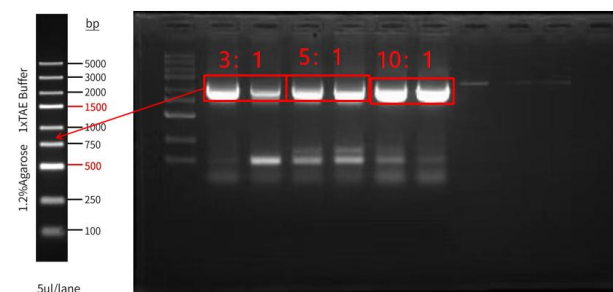
DL2000 DNA Marker



The reagents were added incorrectly during gel extraction, so error-prone PCR was repeated.

Second gel extraction

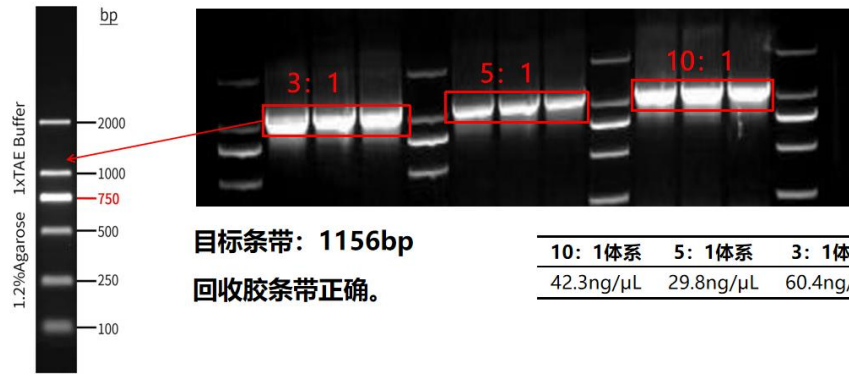
DL5000 DNA Marker



10: 1体系	5: 1体系	3: 1体系
84.2ng/μL	110.9ng/μL	98.8ng/μL

Target band: 1156 bp. The gel- extracted band ranged from 750 to 1000 bp.

Third gel extraction



2.Connection

Connecting the two fragments using the Golden-Gate method.

Goden-gate system (10μL)	10: 1	
5:1		
template	10ng	10ng
10×T4 buffer	1 μL	1 μL
BsaI	0.5 μL	0.5 μL
T4 DNA ligase	0.3 μL	0.3 μL
rCutSmart (10×)	1μL	1μL
vector	10ng	10ng
DDW	5.2μL	4.2 μL

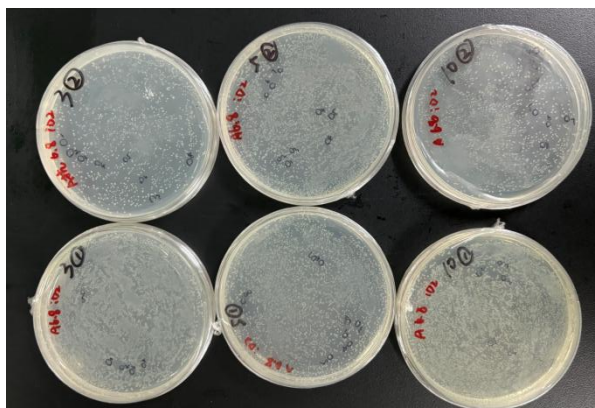
Connection		
temperature	time	cycles
37°C	30min	
37°C	3min	25 X
16°C	3min	25 X
16°C	20min	
37°C	20min	
55°C	15min	
80°C	15min	
25°C	∞	

3.Transformation

The ligated system was transferred into DH5α receptor cells, coated with plates, and cultured inverted at 37°C for 12h;

4.Plating results

Both recovered products were ligated with the vector and transformed, and colonies grew on plates. The library capacities were relatively high, though smearing was observed on some plates.



4.Colony PCR

PBF/PBR was chosen as the primer with a theoretical length of 1508bp. Colony PCR yielded correct target bands, indicating successful ligation.

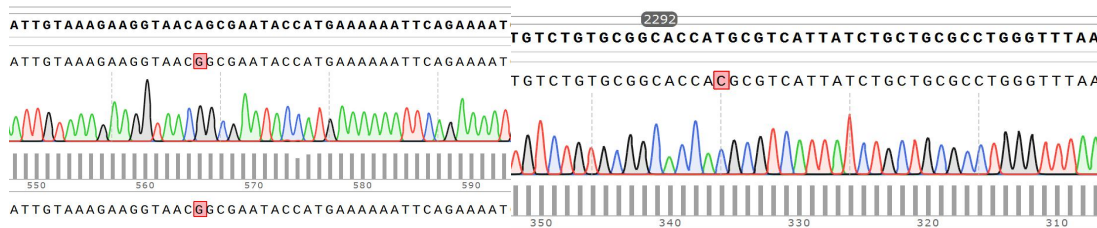


5.Plasmid extraction + send for sequencing

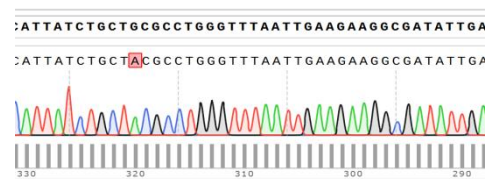
The correctly digested protoplasmid was sent for testing.

3:1

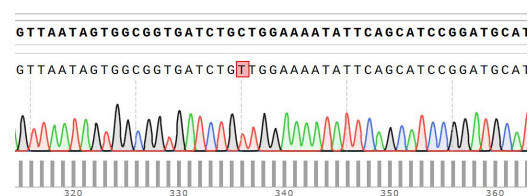
Plasmid1



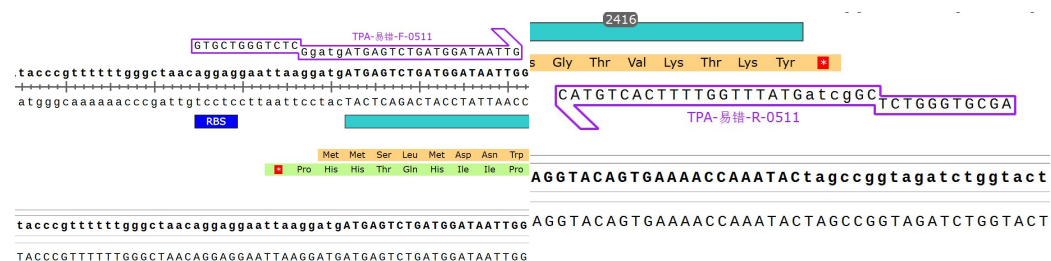
Plasmid2



5: 1



Insertions and deletions of erroneous bases at both termini were eliminated :



10:1 system: no valid mutations.

(B) Fourth- round error- prone PCR amplification of ODM gene

$Mg^{2+} : Mn^{2+} = 10:1$

PCR system (50 μ L)	
template	1ng
F	2ul
R	2ul
10 mM Mg^{2+} solution	1.5 μ L
1 mM Mn^{2+} solution	1.5 μ L
Green Mix	25 μ L
DDW	17 μ L

$Mg^{2+} : Mn^{2+} = 5:1$

PCR system (50 μ L)	
template	1ng
F	2ul
R	2ul
10 mM Mg^{2+} solution	1.5 μ L
1 mM Mn^{2+} solution	3 μ L
Green Mix	25 μ L
DDW	15.5 μ L

Module 3:

(A)Co-transformation of the ABY plasmid and BDDE into BW competent cells, plating on AS resistance plates

1.Co-transformation.

The ABY plasmid (pLB1s-*VanAB-aroY*) and the BDDE plasmid (pYB1a-*dmpB-dmpD-mhpD-mhpE*) were co-transformed into BW competent cells.

2. Plating on AS resistance plates.

The transformation product was plated on AS (ampicillin/streptomycin) resistance plates and cultivated at 37°C for 12 hours.

3.Colony PCR verification

Colony PCR system (10 μ L):

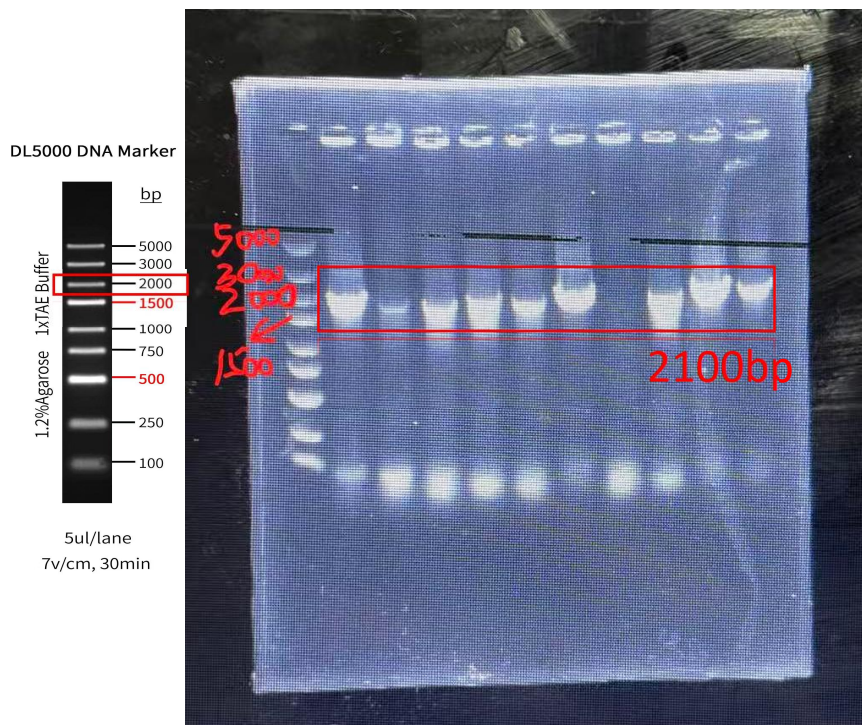
Green Mix	5 μ L
Primer-F	0.4 μ L
Primer-R	0.4 μ L
DDW	4.2 μ L
Colony	—

PCR program:

98°C	5min	} X30
98°C	30s	
57°C	30s	
72°C	2min	
72°C	5min	
25°C	∞	

Single colonies were picked as templates for colony PCR, and the PCR products were verified by agarose gel electrophoresis.

The positive PCR products were excised and recovered from the gel.



(B) Validation of single-gene expression by protein analysis

1. Induction.

50μL of bacterial fluid was taken and transferred to the ZY induction system, and cultivated in a shaking incubator at 30°C for 16 hours. The induction system is as follows:

ZY induction system:

Component	Volume
ZY	4.8mL
50×5052 supplement	100μL
50×M supplement	100μL
MgSO ₄	10μL
1000× trace element	10μL
Bacterial suspension	50μL
arabinose	50μL
Antibiotic	5μL

2. Sample Preparation

The cultures were collected, and the bacterial suspension with a OD₆₀₀=6 unit was measured to obtain the final sample. After centrifugation at 10,000 g for 10 minutes, the supernatant was discarded, and the precipitate was resuspended in 600 μL sterile water. The cells were lysed using an ultrasonic cell disruptor. The lysate was then centrifuged at 10,000 g for 10 minutes to separate the supernatant and sediment. Finally, the precipitate was resuspended in 600 μL sterile water. This procedure completed the sample preparation.

3. point sample

Prepare the protein buffer solution according to the table below. Place the gel in the electrophoresis instrument. After checking for leaks with the buffer solution, mix 16 μL of prepared sample with 4 μL loading buffer. Transfer the mixed sample into the PCR instrument and heat at 100°C for 5 minutes to denature proteins. After cooling, load 16 μL of the mixture into the gel well and add 4 μL of protein marker. The sample loading is now complete.

Concentrating Gel:

10% Lower Gel	2 Gel Units	2 Gel Units	4 Gel Units	6 Gel Units	6 Gel Units	8 Gel Units	10 Gel Units
Solution Components (mL)	10	15	20	20	90	40	30
Water	2.7	5.7	5.5	8.2	8.9	13.2	16.7
30% Acrylamide Solution	4	3.3	6.7	10.0	1.2	13.3	16.7
Tris (pH 8.8)	2.5	1.25	3	3.75	5.0	5.0	6.25
10% SDS	0.1	0.05	0.1	0.15	0.2	0.2	0.25
10% Ammonium	0.1	0.05	0.2	0.15	0.3	0.4	0.25
TEMED	0.002	0.002	0.004	0.006	0.012	0.008	0.01

separation gel:

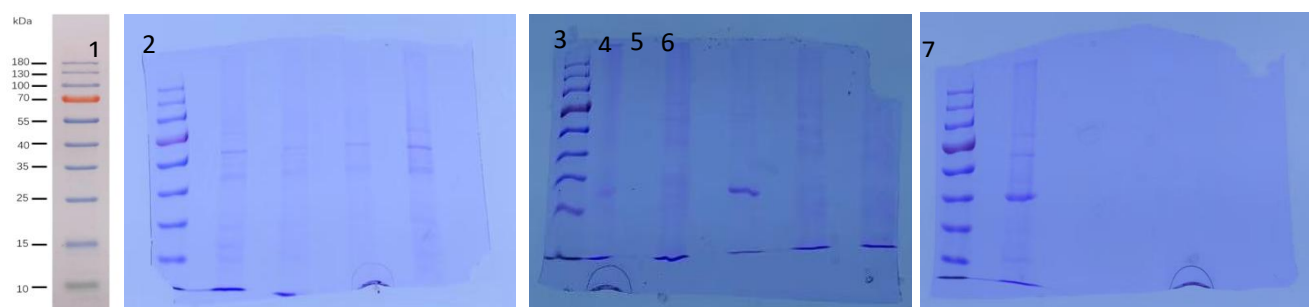
5% Stacking Gel	2 Gel Units	3 Gel Units	4 Gel Units	5 Gel Units	6 Gel Units	8 Gel Units	10 Gel Units	12 Gel Units
Solution Components (mL)	2	3	4	5	6	8	10	12
ddH ₂ O	1.4	2.1	2.7	3.4	4.1	5.5	6.8	8.1
30% Acrylamide Solution	0.33	0.5	0.67	0.83	1	1.3	1.7	2.01
Tris (pH 8.8)	0.25	0.38	0.5	0.63	0.75	1	1.25	1.5
10% SDS	0.02	0.03	0.04	0.05	0.06	0.08	0.1	0.12
10% Ammonium Persulfate (APS)	0.02	0.03	0.04	0.05	0.06	0.08	0.1	0.12
TEMED	0.002	0.003	0.004	0.005	0.006	0.008	0.01	0.012

4.electrophoresis

Connect the power supply and run the gel at 80 V for 30 minutes, then run it at 120 V for about 1 hour

5.Dyeing and decoloring

Gently remove the gel from the water, then place it in the staining solution. Boil the solution and incubate the gel with heated staining liquid on a small shaker at room temperature for approximately 45 minutes. Rinse the gel thoroughly with distilled water to remove residual staining. Add protein decolorizing solution, heat it to boiling, and decolorize the heated solution on a small shaker at room temperature for 30 minutes. The results are shown in the figures:



- 1: supernatant of uninduced empty vector
- 2: empty vector
- 3: supernatant of induced dmpB
- 4: pellet of induced dmpB
- 5: supernatant of uninduced dmpB
- 6: pellet of uninduced dmpB
- 7: pellet of induced mhpE

Week 5 & 6: 6.12-6.25

Module 1:

1. Inoculation

Original strains: BW-pYB1a-*vdh*、BW-pYB1a-*nox*、BW25113、BW-pYB1a-*eGFP*、BW-pYB1a-*vdh-nox*

Cultivation in a shaker at 37°C for 12 hours

2. Induction

Induction of BW-pYB1a-*vdh*、BW-pYB1a-*nox*、BW25113、BW-pYB1a-*eGFP*、BW-pYB1a-*vdh-nox*

Induction System Component	Volume
ZY	4.8 mL
50x 5052 supplement	100 μ L
50x M supplement	100 μ L
MgSO ₄	10 μ L

1000x trace element	10 μ L
Bacterial suspension	50 μ L
arabinose	50 μ L
Antibiotic	5 μ L

Cultivation in a shaker at 30°C for 16 hours.

3. protein detection

1. Sample Preparation

The pYB1a-Vdh-Nox、pYB1a-Vdh Protein strains were cultured under induction conditions at 30°C for 16 hours. The cultures were collected, and the bacterial suspension with a OD₆₀₀ = 6 unit was measured to obtain the final sample. After centrifugation at 10,000 g for 10 minutes, the supernatant was discarded, and the precipitate was resuspended in 600 μ L sterile water. The cells were lysed using an ultrasonic cell disruptor. The lysate was then centrifuged at 10,000 g for 10 minutes to separate the supernatant and sediment. Finally, the precipitate was resuspended in 600 μ L sterile water. This procedure completed the sample preparation.

2. point sample

Prepare the protein buffer solution according to the table below. Place the gel in the electrophoresis instrument. Transfer the sample into the PCR instrument and heat at 100°C for 10 minutes to denature proteins. After checking for leaks with the buffer solution, mix 32 μ L of prepared sample with 8 μ L loading buffer. After that, load 12 μ L of the mixture into the gel well and add 10 μ L of protein marker. The sample loading is now complete.

Concentrating Gel:

12% Lower Gel	2 Gel Units	3 Gel Units	4 Gel Units	5 Gel Units	6 Gel Units	8 Gel Units	10 Gel Units
Solution Components (mL)	10	15	20	25	30	40	50
Water	3.3	4.9	6.6	8.2	9.9	13.2	16.5
30% Acrylamide Solution	4	6	8	10	12	16	20
Tris (pH 8.8)	2.5	3.8	5	6.3	7.5	10	12.5
10% SDS	0.1	0.15	0.2	0.25	0.3	0.4	0.5
10% Ammonium Persulfate	0.1	0.15	0.2	0.25	0.3	0.4	0.5
TEMED	0.004	0.006	0.008	0.01	0.012	0.016	0.02

separation gel:

5% Stacking Gel	2 Gel Units	3 Gel Units	4 Gel Units	5 Gel Units	6 Gel Units	8 Gel Units	10 Gel Units	12 Gel Units
Solution Components (mL)	2	3	4	5	6	8	10	12
ddH ₂ O	1.4	2.1	2.7	3.4	4.1	5.5	6.8	8.1
30% Acrylamide Solution	0.33	0.5	0.67	0.83	1	1.3	1.7	2.01
Tris (pH 8.8)	0.25	0.38	0.5	0.63	0.75	1	1.25	1.5
10% SDS	0.02	0.03	0.04	0.05	0.06	0.08	0.1	0.12
10% Ammonium Persulfate (APS)	0.02	0.03	0.04	0.05	0.06	0.08	0.1	0.12
TEMED	0.002	0.003	0.004	0.005	0.006	0.008	0.01	0.012

3. electrophoresis

Connect the power supply and run the gel at 80 V for 30 minutes, then run it at 120 V for about 1 hour

4. Dyeing and decoloring

Gently remove the gel from the water, then place it in the staining solution. Boil the solution and incubate the gel with heated staining liquid on a small shaker at room

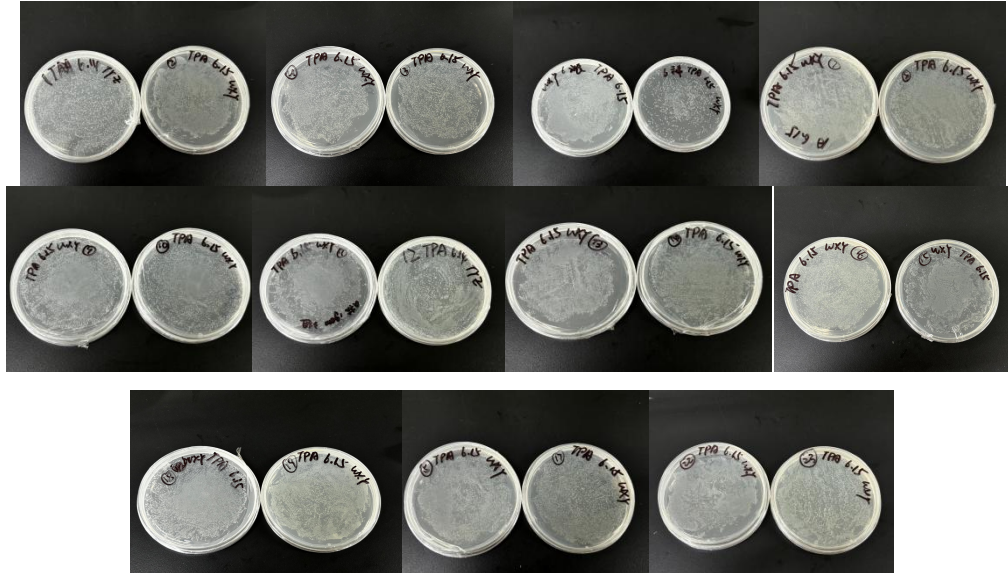
temperature for approximately 45 minutes. Rinse the gel thoroughly with distilled water to remove residual staining. Add protein decolorizing solution, heat it to boiling, and decolorize the heated solution on a small shaker at room temperature for 30 minutes. The results were failure.

Module 2:

(A) Extract the mixed plasmid of pYB1a- ODM.

1. Transformation and plating of pYB1a- ODM yielded a high library capacity.

Twenty-two tubes were assembled by Golden Gate assembly and transformed.



Library capacity: 1.61×10^5 CFU

2. Scrape the colonies and extract plasmids.

Colony scraping :

Retrieve the colony plates incubated at 37 °C for 12 h. Perform all operations in a laminar flow hood. Add 5 mL antibiotic-containing LB liquid medium to each plate, and discard the remaining LB ;

Scrape all colonies completely from each plate using a sterile disposable inoculating loop to fully suspend the bacterial cells in the liquid ;

Transfer all bacterial suspensions eluted from the plates into an empty conical flask with a pipette ;

Seal the conical flask and incubate with shaking at 37 °C for 4–5 h in a shaker. Store the bacterial culture after incubation.

Extract mixed plasmids :

Prepare 24 capped 2 mL centrifuge tubes (corresponding to 24 plates) and fill them with bacterial suspension ;

Centrifuge each tube twice: discard the supernatant after each run, replenish bacterial suspension and re-centrifuge.

Add reagents to each tube following standard protocols ;

Perform inversion mixing, lysis, neutralization and centrifugation according to routine plasmid extraction procedures to collect supernatants ;

Load the supernatant from each 2 mL tube onto one spin column, followed by washing and elution. Twenty- four eluate fractions were obtained.

3.Transformation

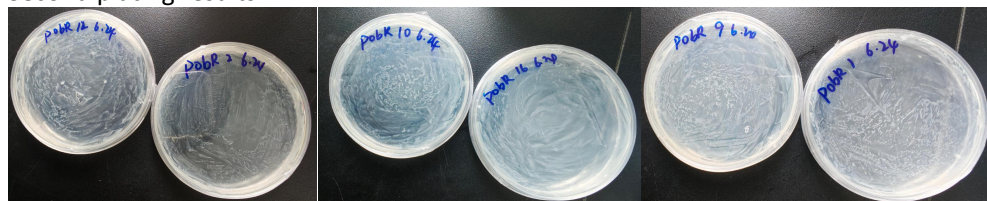
The mixed plasmid was diluted to 42 ng/ μ L. Twenty- five aliquots of pobR competent cells were taken, and 1 μ L mixed plasmid was added to each. The mixtures were incubated on ice for 25 min, heat-shocked at 42 $^{\circ}$ C for 45 s, and then placed on ice for another 2–3 min. After incubation, 600 μ L LB was added to each EP tube, followed by shaking incubation at 37 $^{\circ}$ C for 45 min. The samples were centrifuged at 4200 rpm for 2 min. In a laminar flow hood, 100 μ L of the suspension was pipetted, mixed and plated. One plate was prepared as a diluted plate.

First plating results:



It was considered that the plates were not dried sufficiently after plating, so transformation was repeated.

Second plating results:



The high colony density was attributed to either excessive bacterial concentration or antibiotic inactivation.

Module 3:

(A)Add different concentrations of sodium pyruvate to measure the sensor growth rate

1.Groups with different concentrations of sodium pyruvate.

Different concentrations of sodium pyruvate were added (with a no-substrate control group). The sensor culture was inoculated into the 96-well plate and the OD600 was continuously monitored with the microplate reader (48 hours).

2.Prepare the pyruvate sensor competent cells

1.Preparation of the pyruvate sensor competent cells.

The competent cells of the pyruvate sensor strain were prepared for subsequent co-transformation.

3.Pick colonies and shake culture after co-transformation

After the co-transformation and plating, single colonies were picked, inoculated into 5mL of LB medium, and cultivated in a shaking incubator at 37°C for 12 hours.

(B)Validation of single-gene expression by protein analysis

1.Induction.

50μL of bacterial fluid was taken and transferred to the ZY induction system, and cultivated in a shaking incubator at 30°C for 16 hours. The induction system is as follows:

ZY induction system:

Component	Volume
ZY	4.8mL
50×5052 supplement	100μL
50×M supplement	100μL
MgSO4	10μL
1000× trace element	10μL
Bacterial suspension	50μL
arabinose	50μL
Antibiotic	5μL

2.Sample Preparation

The cultures were collected, and the bacterial suspension with a OD₆₀₀=6 unit was measured to obtain the final sample. After centrifugation at 10,000 g for 10 minutes, the supernatant was discarded, and the precipitate was resuspended in 600 μL sterile water. The cells were lysed using an ultrasonic cell disruptor. The lysate was then centrifuged at 10,000 g for 10 minutes to separate the supernatant and sediment. Finally, the precipitate was resuspended in 600 μL sterile water. This procedure completed the sample preparation.

3.Point sample

Prepare the protein buffer solution according to the table below. Place the gel in the electrophoresis instrument. After checking for leaks with the buffer solution, mix 16 μL of prepared sample with 4 μL loading buffer. Transfer the mixed sample into the PCR instrument and heat at 100°C for 5 minutes to denature proteins. After cooling, load 16 μL of the mixture into the gel well and add 4 μL of protein marker. The sample loading is now complete.

Concentrating Gel:

10% Lower Gel	2 Gel Units	2 Gel Units	4 Gel Units	6 Gel Units	6 Gel Units	8 Gel Units	10 Gel Units
Solution Components (mL)	10	15	20	20	90	40	30
Water	2.7	5.7	5.5	8.2	8.9	13.2	16.7
30% Acrylamide Solution	4	3.3	6.7	10.0	1.2	13.3	16.7
Tris (pH 8.8)	2.5	1.25	3	3.75	5.0	5.0	6.25
10% SDS	0.1	0.05	0.1	0.15	0.2	0.2	0.25
10% Ammonium	0.1	0.05	0.2	0.15	0.3	0.4	0.25
TEMED	0.002	0.002	0.004	0.006	0.012	0.008	0.01

separation gel:

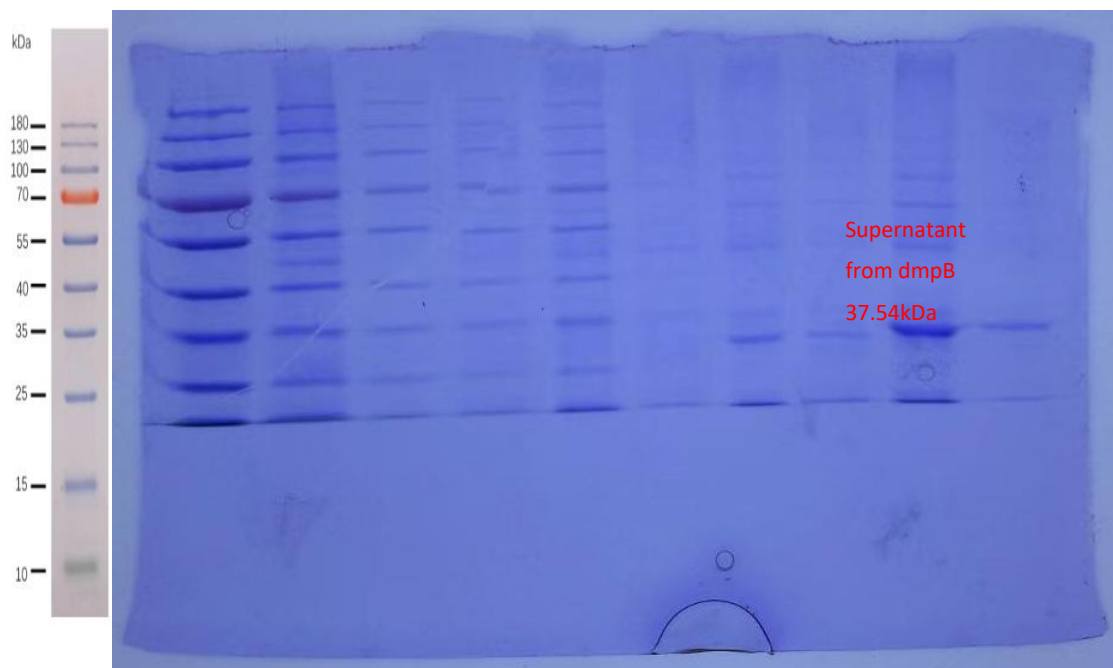
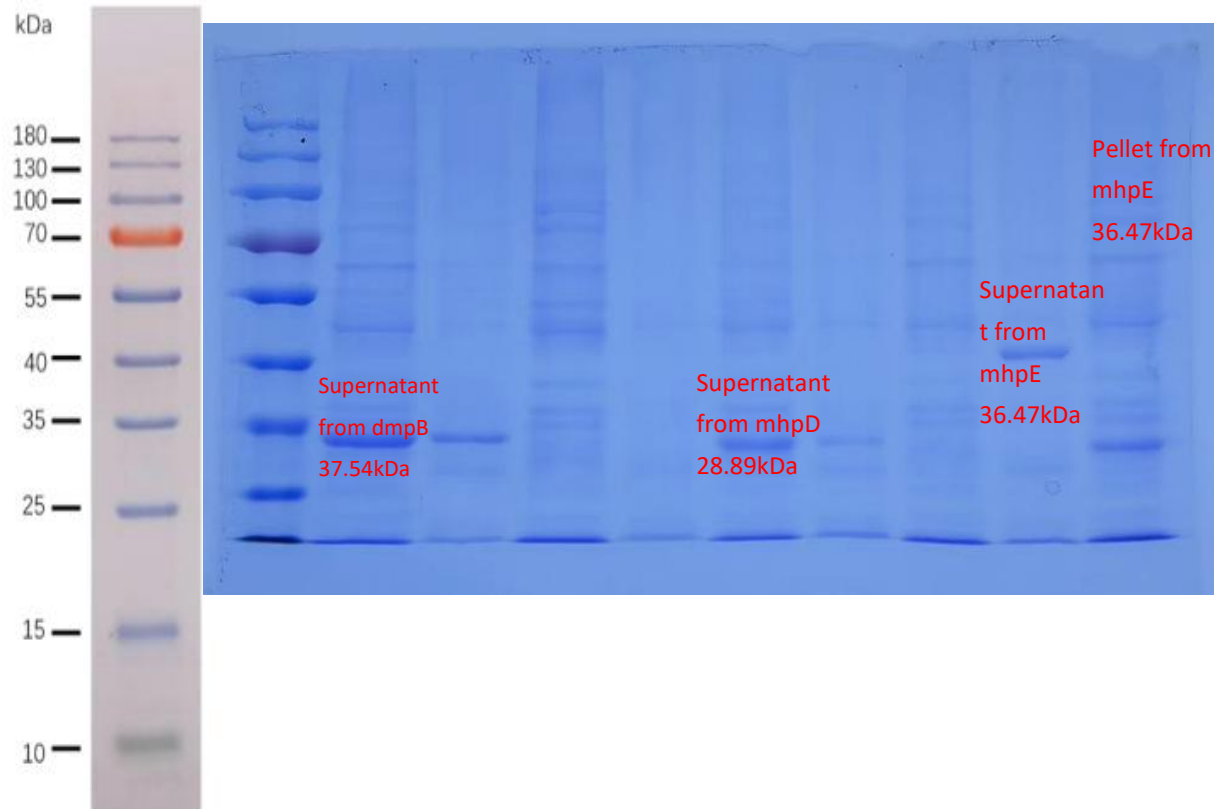
5% Stacking Gel	2 Gel Units	3 Gel Units	4 Gel Units	5 Gel Units	6 Gel Units	8 Gel Units	10 Gel Units	12 Gel Units
Solution Components (mL)	2	3	4	5	6	8	10	12
ddH ₂ O	1.4	2.1	2.7	3.4	4.1	5.5	6.8	8.1
30% Acrylamide Solution	0.33	0.5	0.67	0.83	1	1.3	1.7	2.01
Tris (pH 8.8)	0.25	0.38	0.5	0.63	0.75	1	1.25	1.5
10% SDS	0.02	0.03	0.04	0.05	0.06	0.08	0.1	0.12
10% Ammonium Persulfate (APS)	0.02	0.03	0.04	0.05	0.06	0.08	0.1	0.12
TEMED	0.002	0.003	0.004	0.005	0.006	0.008	0.01	0.012

4. Electrophoresis

Connect the power supply and run the gel at 80 V for 30 minutes, then run it at 120 V for about 1 hour

5. Dyeing and decoloring

Gently remove the gel from the water, then place it in the staining solution. Boil the solution and incubate the gel with heated staining liquid on a small shaker at room temperature for approximately 45 minutes. Rinse the gel thoroughly with distilled water to remove residual staining. Add protein decolorizing solution, heat it to boiling, and decolorize the heated solution on a small shaker at room temperature for 30 minutes. The results are shown in the figures:



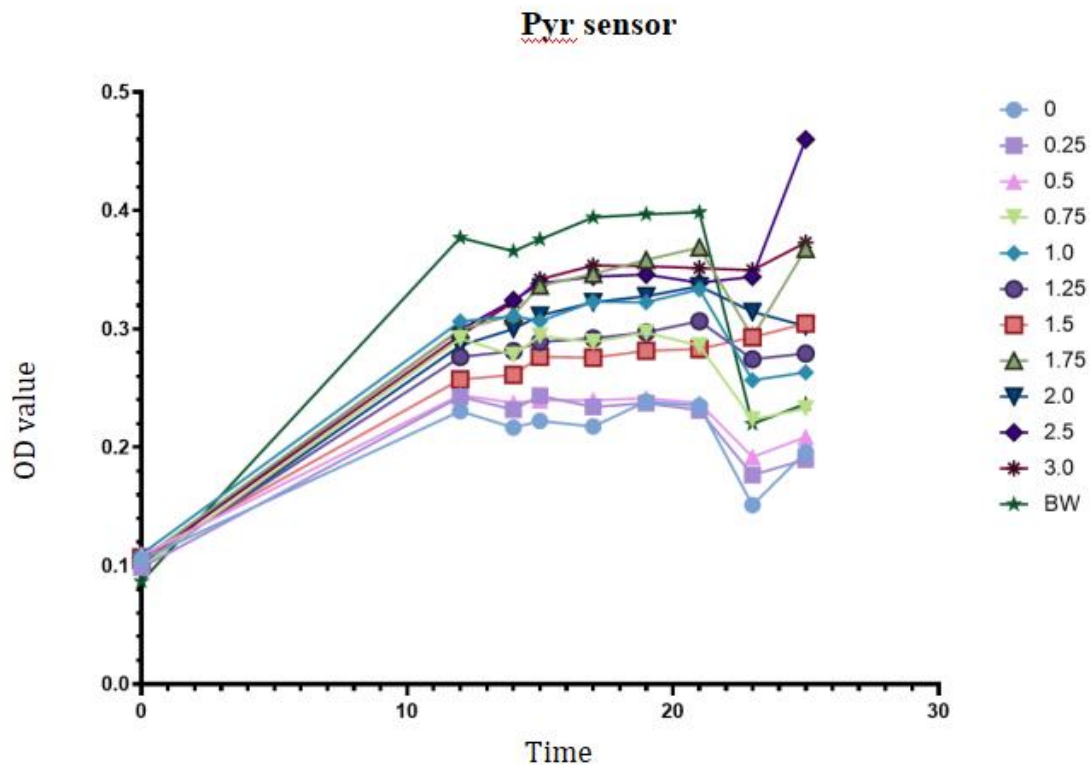
dmpB: 37.54kDa, dmpD: 31.68kDa, mhpD: 28.89kDa, mhpE: 36.47kDa

(C) Growth assay with exogenous sodium pyruvate

Different concentrations of exogenous sodium pyruvate were added to 1×M9 medium (inoculation volume 0.002%), and the growth rate of the sensor was recorded with the microplate reader.

1. Induction and OD monitoring

The strain was induced, and the OD value was measured with the microplate reader every 2 hours; the data were recorded.



2. Results (growth curves of BW-ΔgldA-ΔmaeB-ΔpykF at 12 h and 23 h, inoculation volume 0.002%):

1. The growth rate of the pyruvate sensor was significantly lower than that of the BW parent strain, indicating that the sensor responds to pyruvate limitation.
2. Overall trend: when the sodium pyruvate concentration was below 2.0 g/L, the growth rate of the sensor increased with increasing pyruvate concentration.
3. At 23 h, a sharp decrease in OD was observed and was judged to be an operational error.
4. According to the data, problems existed in the process operation; the characterization should be repeated.

(C) Full-pathway verification using the pyruvate sensor

Objective: to verify whether the complete vanillic acid-to-pyruvate pathway functions in the co-transformed strain (pyruvate sensor + pLB1s-*VanAB-aroY* + pYB1a-*dmpBD-mhpDE*). If the pathway works, vanillic acid is converted to pyruvate, and the growth of the pyruvate sensor should be restored.

1. Colony picking, shake culture and induction

Single colonies of the co-transformed pyruvate sensor were picked, inoculated and induced.

2. Substrate and cofactor addition

1 mM vanillic acid and 2 mM NADH were added to the induced cultures in 1×M9 medium.

3. OD monitoring

The OD value was measured every 2 hours and compared with the BW parent strain and the pyruvate sensor without the pathway.

The group design is shown in Table 3.

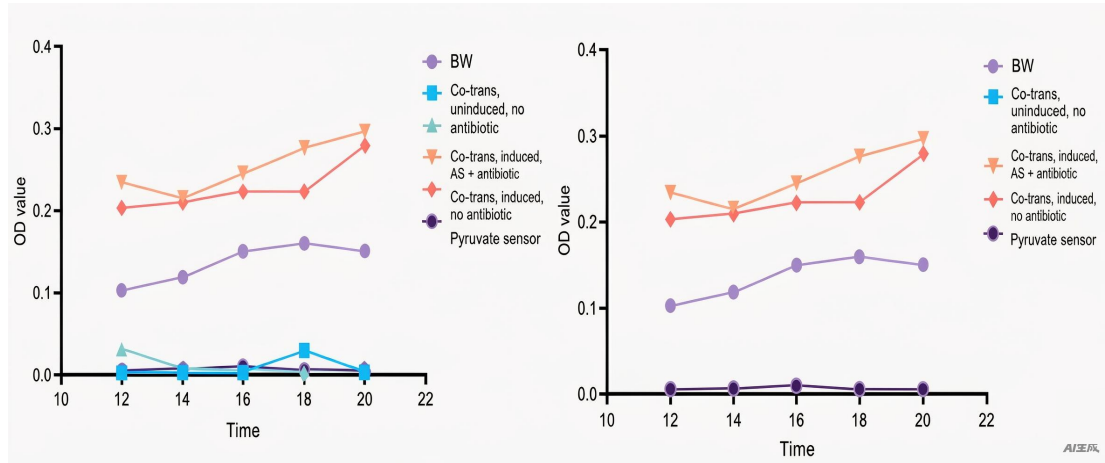
Table 3 Addition system for full-pathway verification

1×M9 medium preparation:

1×M9 base solution(30mL)				
5xM9 salt solution	6mL			
20% glycerol	3mL			
1M MgSO ₄	60μL			
streptomycin (S) resistance	30μL			
Ara	300μL			
1M CaCl ₂	3μL			
DDW	20.607mL			

Group	Medium	Inoculum (50 μL)	Additives	Note
BW	Antibiotic-free 1×M9	BW culture	—	Control
Co-transformed × No-Ab (uninduced)	Antibiotic-free 1×M9	Uninduced co-transformed culture	—	No growth
Co-transformed × AS (uninduced)	1×M9 with AS antibiotics	Uninduced co-transformed culture	—	No growth
Co-transformed √ No-Ab (induced)	Antibiotic-free 1×M9	Induced co-transformed culture	1 mM vanillic acid, 2 mM NADH	—
Co-transformed √ AS (induced)	1×M9 with AS antibiotics	Induced co-transformed culture	1 mM vanillic acid, 2 mM NADH	—
Pyruvate sensor	Antibiotic-free 1×M9	Pyruvate sensor culture	—	Control

All groups were cultivated at 30 °C with shaking; the OD value was measured every 2 hours.



Results and analysis:

1. The growth rate of the pyruvate sensor was significantly lower than that of BW, further confirming the validity of the sensor.
2. Because of the variable-setting problems, the effectiveness of the full pathway could not be conclusively determined from the current data.
3. Overall trend: after the addition of vanillic acid and NADH, the growth rate of the induced co-transformed group approached that of the BW parent strain, preliminarily indicating that the pathway is effective.
4. Improvement suggestions: (i) add an induced group without vanillic acid/NADH and an uninduced group with vanillic acid/NADH for comparison; (ii) the antibiotic-free groups can be omitted; (iii) record the initial OD values to strengthen the evidence.